

Microrobotic Locomotion on Water Surfaces via Onboard Dielectric Barrier Discharge

Yuhang Song

College of Integrated Circuit Science
and Engineering
Nanjing University of Posts and
Telecommunications
Nanjing, China
syh_SML_NJUPT@163.com

Hongji Hui

College of Integrated Circuit Science
and Engineering
Nanjing University of Posts and
Telecommunications
Nanjing, China
ichuihongji@163.com

Qiannan Tao

College of Integrated Circuit Science
and Engineering
Nanjing University of Posts and
Telecommunications
Nanjing, China
qntao_lea@njupt.edu.cn

Wei Li

College of Integrated Circuit Science
and Engineering
Nanjing University of Posts and
Telecommunications
Nanjing, China
wei.li@njupt.edu.cn

Abstract—Dielectric barrier discharge (DBD) converts electrical energy into mechanical energy, generating airflow and producing thrust. As a novel power source, DBD systems offer advantages such as low noise, no moving parts, and a compact form factor. These characteristics make them particularly suitable for application in small mechanical devices. In this study, we conduct thrust tests and Lissajous figure analysis on DBD thrusters based on different materials and their thickness, and structures, together with the input signal. Next, we develop the first micro-surface vessel powered by a DBD thruster, successfully achieving navigation on water. The experimental results indicated that the thrust performance of polyether ether ketone (PEEK) surpassed that of polyimide (PI) and FR-4 epoxy glass fiber laminate (FR-4). The optimal structural thickness of the PEEK material is 0.1 mm, with an electrode spacing of 4 mm. This structure can generate a thrust of 0.29 mN at a peak voltage of 15.3 kV, sufficient to propel a surface vessel. The power consumption at this thrust level is 2.15 W. These research findings highlight the potential of DBD thrusters as power systems for micro-mechanical devices.

Keywords—Dielectric barrier discharge, Surface vessel, Electro-hydrodynamic propulsion, Lissajous figure

I. INTRODUCTION

Electrohydrodynamics (EHD) has garnered significant attention in both fundamental research and practical applications following the discovery that corona discharge induced by a strong electric field can generate an ionized airflow. The Coulomb force exerted by the electric field accelerates the ionized particles relative to neutral ones. These ions subsequently collide with neutral particles, transferring momentum and generating an ion wind. A recoil force is simultaneously produced, acting in the opposite direction to the ion acceleration. Since its initial discovery, this

mechanism has been progressively refined and more accurately characterized [1].

In recent years, EHD has become a popular area of research, with DBD emerging as a favored option among active flow control methods [2,3]. This technique has several unique advantages, including fully electrical operation, an absence of moving parts, a lightweight design, a rapid response time, and ease of installation on diverse surfaces without causing damage. These characteristics set DBD plasma actuators apart from other active propulsion strategies. DBD operates on the principle of creating a plasma discharge between two electrodes separated by an insulating dielectric material. When a high alternating voltage is applied across the electrodes, it ionizes the gas in the gap between them, leading to the formation of a plasma region. DBD can serve dual purposes, acting both as an ozone generator for sterilization and disinfection [4,5] and as a force-producing actuator. The DBD actuator has been successfully employed in various electromechanical systems. For example, DBD actuators are utilized in jet systems to replace the fuel-powered engines [6-8], and are also applied in devices like ailerons to substitute mechanical moving parts to some extent to achieve flow control [9,10].

The performance of a DBD actuator is influenced by various parameters, both internal and external. Internal factors include materials [11,12], electrode structures [13], and the geometry of the overall design [12-15]. External factors often refer to the form of the input signal, encompassing waveform [15], amplitude [12,15,17], and frequency [12,18]. Additionally, temperature, humidity [19,20], and the surrounding gaseous environment [21] can also influence performance. Apart from practical experiments, theoretical simulations [22-25], have also revealed some characteristics of DBD. It is indicated that vortex formation plays a critical role

in DBD performance [26], with the input signal affecting vortex dynamics [27]. To assess these parameters, various measurement techniques have been employed, such as thrust measurement with a balance [12,18] and Lissajous figures measurement for power consumption [28]. The inherent design features of DBD actuators, together with flexible parameter tuning, allow for near-arbitrary geometries while maintaining compactness and high energy efficiency.

Despite the numerous advantages and widespread applications of DBD structures, their relatively low thrust output has resulted in limited research focused on their use in propulsion applications. However, the compact nature of DBD structures offers distinct advantages for propelling small-scale mechanical devices [29,30]. In our work, we first fabricated and experimentally tested a DBD actuator to develop an efficient thruster design. The experimental results demonstrate that, by carefully tuning the DBD actuator, thrust can be enhanced while power consumption is reduced. Building upon this foundation, we designed a surface vessel assembled with a DBD actuator and evaluated its forward propulsion performance. The experiments confirmed the feasibility and effectiveness of using the DBD actuator as the primary thruster. The vessel operates with low noise characteristics, highlighting its potential for sustainable applications.

II. MECHANISM AND FABRICATION

A. Mechanism

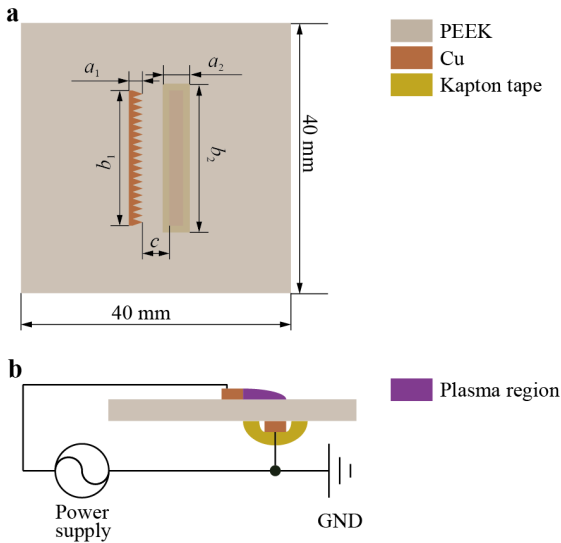


Fig. 1. Schematic of DBD actuator: (a) Top-down view. (b) Side view.

The DBD actuator serves as the driving force for the vessel, and in this study, it is treated as an independent component to be specifically designed and optimized. A typical DBD actuator setup is shown in Fig. 1a, which represents the basic surface DBD (SDBD) configuration, and it is the same model applied in the vessel. This typical SDBD system consists of three major components: the dielectric layer, the emitter, and the collector. The dielectric layer serves as the base of DBD and acts as an electrical isolator between the electrodes. The emitter is the exposed electrode, which is connected to the HV side of the AC power system, while the

collector is the encapsulated one, isolated with an electric isolator like Kapton tape, and is connected to the ground. Together with the dielectric layer, the two electrodes form the working part of the DBD actuator. The plasma region, which generates the electric thrust, typically appears on the emitter side, as shown in Fig. 1b.

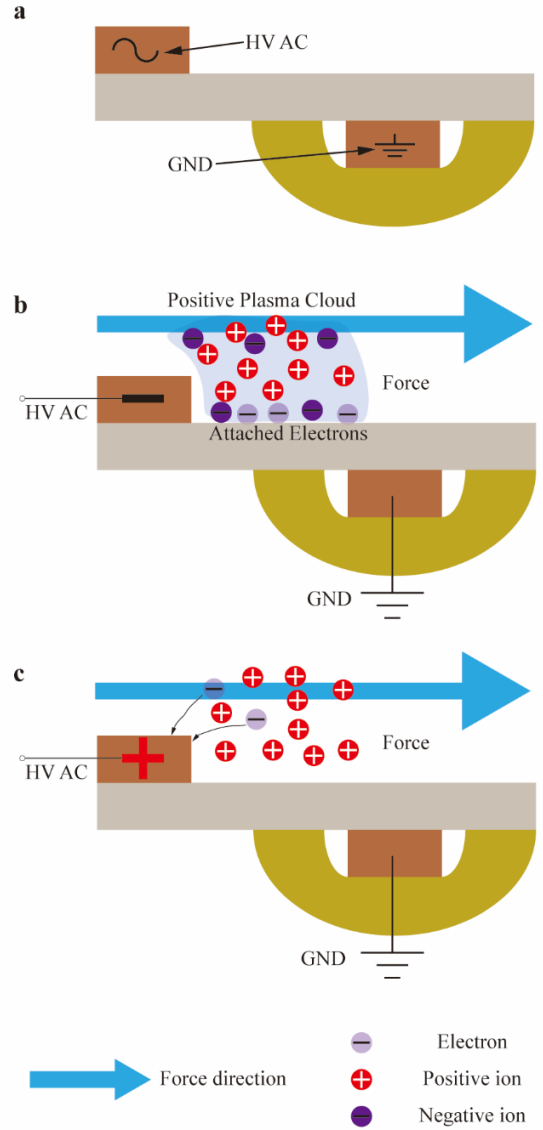


Fig. 2. Schematic illustrating the principle of the DBD actuator: (a) Diagram of the working DBD actuator during the positive-half cycle of the input signal. (b) Diagram of the working DBD actuator during the negative-half cycle of the input signal.

The working principle of the DBD actuator is shown in Fig. 2. As shown in Fig. 2a, A high-voltage AC signal is periodically applied to the electrodes, with the emitter receiving the AC voltage while the collector remains at zero potential. The AC voltage can take the form of a sine wave, square wave, triangle wave, or other waveforms. During the negative half-cycle (Fig. 2b), the emitter induces a strong electric field, which ionizes the surrounding neutral particles.

Electrons are generated at the emitter due to secondary emission, with the strong electric field causing electron avalanche. The electrons are separated from the positive ions and attach to the dielectric surface, leading to the charging of the dielectric surface by the attached electrons. The remaining positive ions form a positive plasma cloud, and some electrons combine with oxygen atoms to form negative ions. These negative ions drive neutral molecules, inducing gas motion away from the exposed electrode. Although the movement of the positive ions towards the exposed electrode during the negative half-cycle generates a force directed towards the electrode, for millimeter-scale electrode DBD devices, the integrated force is directed away from the exposed electrode. In low-frequency conditions, by the end of the negative half-cycle, nearly all electrons on the dielectric surface disappear through recombination with ions. As shown in Fig. 2c, the plasma around the positively exposed electrode exists solely in the form of a positive corona. Electrons generated in the strong electric field drift towards the exposed positive electrode. These positive ions move along the dielectric surface in the opposite direction, imparting momentum to the gas. Therefore, the direction of the gas's momentum remains the same as during the negative half-cycle [31].

B. Fabrication

The DBD actuators in this study are primarily fabricated using sputtering, laser cutting, and manual assembly techniques. The fabrication process is illustrated in Fig. 3.

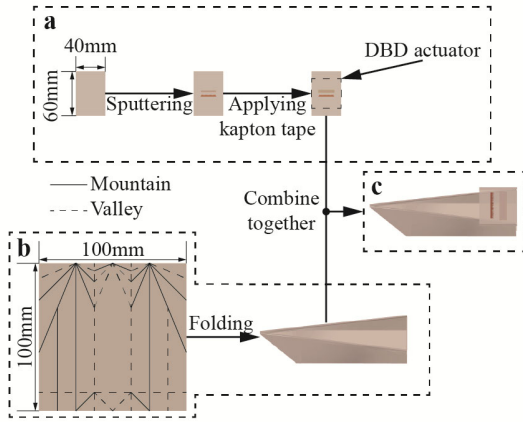


Fig. 3. Manufacturing process of the surface vessel with the DBD actuator. (a) Fabrication process of DBD actuator. (b) Fabrication process of surface vessel. (c) Assembly process of the DBD actuator and the surface vessel.

Laser cutting is employed to obtain the dielectric layer base and a simple mask for sputtering, while sputtering is used to deposit electrodes onto the dielectric layer. This step can also be simplified by using copper tape as the electrode. Finally, manual assembly is used to isolate the collector with Kapton tape. Inspired by origami [32,33], we design and fabricate a surface vessel to achieve three-dimensional (3D) microrobot geometries from the two-dimensional surface dielectric barrier discharge (2D-SDBD) actuation principle. The main part of the vessel is hand-folded from a sheet of material. The origami design facilitates a high degree of flexibility in the assembly process. The low thickness of the

polyether ether ketone (PEEK) plate used for folding, combined with sputtering for electrode formation, is the primary factor contributing to the vessel's lightweight construction. The fabrication process for the designed vessel involves separately crafting the vessel and actuator before assembling them together. The final step is the wiring to establish electrical connections. The overall design results in a compact and refined vessel capable of steady movement.

III. EXPERIMENT OF PARAMETER ANALYSIS

A. Experiment Setup

The actual design of our DBD actuator is shown in Fig. 1a, which includes several variables and parameters that will be discussed in detail. Many parameters of the DBD actuator influence its performance and power consumption, and among these, some factors play a more significant role, such as the electrode geometry, the type of dielectric layer, and the voltage amplitude. Regarding electrode geometry, this study uses serrated electrodes, as they have been demonstrated to produce stronger thrust output in several research studies [34,35].

The set of parameters we experimented with includes two factors: the material of the dielectric layer and the input voltage amplitude. The standard set for experiments is described as follows. Electrode width for a_1 and a_2 indicated in Fig. 1a is 2 mm, electrode length for b_1 indicated in Fig. 1a was 20 mm, the insulated package length of the collector b_2 indicated in Fig. 1a was 22 mm, and the width of the insulated package of the collector was 1 mm larger on each side compared to the electrode. The gap c between the two electrodes indicated in Fig. 1a was 4 mm. The dielectric layer was 0.1-mm-thickness PEEK. The size of the test samples was 40 mm (length) $\times$ 40 mm (width). The input voltage signal was a 50 Hz sine wave, ranging from 0 to approximately 17 kV. The materials tested included PEEK, polyimide (PI), and FR-4 epoxy glass fiber laminate (FR-4).

Fig. 4a shows the experimental platform for testing thrust and power consumption. The balance is used to measure thrust data with a precision of milligrams. The principle of thrust measurement is illustrated as follows. The DBD actuator sample will generate lift or downforce, which is equivalent to the application or removal of an object from a zeroed balance. This will result in a change to the number displayed on the balance. The average number of the acquired data, multiplied by the gravitational acceleration, will be approximately equivalent to the generated thrust of the DBD actuator under certain conditions.

The other components are used to collect voltage data for both the input and the capacitor. These data can then be used to generate a Lissajous figure (Fig. 4b), which reflects the energy consumption per cycle. The working principle of this figure can be described by the following equation:

$$\bar{P}(t) = f \oint_{\tau} V dQ \quad (1)$$

where $\overline{P}$ is the average power consumption of a cycle, V is the input voltage, and Q is the charge measured through the additional capacitance. T represents the duration of the alternating current's period, and f is the AC frequency.

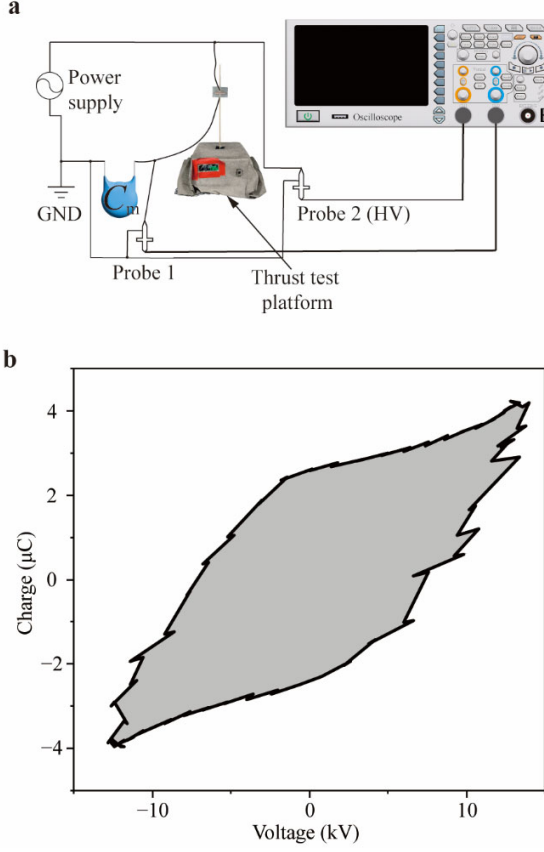


Fig. 4. Diagram of test platform and Lissajous figure: (a) Illustration of test platform for thrust and power. (b) The Lissajous figure under 14.4kV (the area of the shaded part represents the energy consumption of a cycle).

The measurement principle is based on assuming the DBD device behaves as an equivalent capacitor, whose capacitance varies with its operational state. The DBD capacitor essentially exhibits two states: active and inactive. As illustrated in Fig. 4a, two series capacitors are present: C_m and the DBD capacitor (denoted as C_0), with an equivalent capacitance C_e . In the active state, C_0 approaches breakdown, resulting in $C_e \approx C_m$; in the inactive state, the equivalent capacitance corresponds to the series combination of C_m and C_0 . The value of C_e can be determined from the slope of the Lissajous figure. Variations in C_e induced by the input signal cause charge transfer into and out of the DBD. As the DBD is an energy-consuming device, this charge transfer corresponds to energy expenditure. The shaded area in Fig. 4b represents the energy consumed per cycle; hence, the power can be calculated as the product of the operating frequency and the energy consumption per cycle.

B. Experiment Analysis

The performance is primarily studied by analyzing the

normalized thrust and power consumption per centimeter of the DBD actuator. The processed data are presented in Fig. 5.

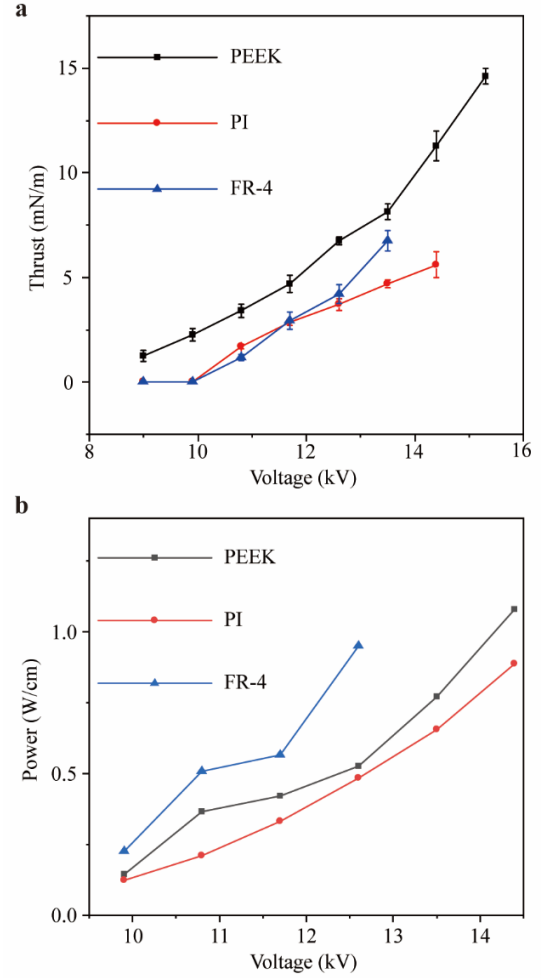


Fig. 5. Experiment results for comparison of materials: (a) Thrust plotted versus voltage comparison for different dielectric layer materials. (b) Power plotted versus voltage comparison for different dielectric layer materials.

It indicates that, for a given configuration (excluding the dielectric material), PEEK outperforms both PI and FR-4. The superiority is reflected in thrust, voltage tolerance, and relatively low power cost. Our subsequent experiments with vessels also demonstrate that PEEK, which achieves similar performance at lower voltages compared to the other materials, maintains stable operation over extended periods. However, the power consumption does not follow a similar trend, as indicated by the data in Fig. 5b. According to these results, PI appears to have the lowest power consumption among the three materials, followed by PEEK, with FR-4 exhibiting the highest power consumption. Based on these findings, PEEK is identified as the optimal material for the vessel's propulsion system.

Regarding the voltage, it has a significant impact on both thrust generation and power consumption. The data indicate that thrust and power are approximately proportional to the square of the applied voltage, implying that higher input voltage amplitudes result in increased output thrust as well as

higher power consumption.

IV. EXPERIMENTS OF THE VESSEL

Based on these experimental findings, 100- μm -thickness PEEK was ultimately selected as the dielectric layer material for the DBD propulsion system of the micro-surface vessel. After installing the DBD thruster on the surface vessel, further practical performance tests were conducted. The fabricated DBD-driven vessel is compact and highly reliable. The experimental setup, as shown in Fig. 6a, involves placing the vessel in an acrylic tank much larger than the vessel's own diameter. Its motion is then observed after applying power.

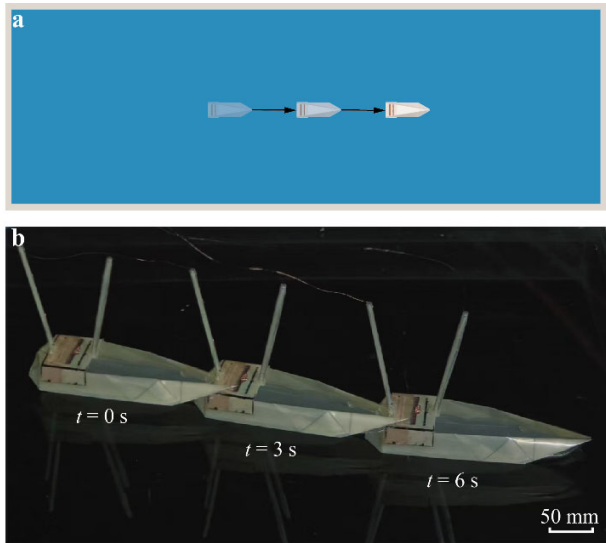


Fig. 6. Demonstration of our surface vessel: (a) Diagram of the movement of the surface vessel in an acrylic tank (from left to right). (b) Demonstration of the DBD-powered surface vessel moving on the water.

The operation of vessel is shown in Fig. 6b, presented in an overlapping multi-frame format. It demonstrated steady and smooth forward motion at an average speed of 0.3 body lengths (8 cm) per second, with a weight of approximately 2 g.

V. CONCLUSION

We demonstrated the first launch of DBD-powered micro-surface vessels. Experimental results revealed that the ideal material is 0.1-mm-thick PEEK, accompanied by an electrode spacing of 4 mm. This configuration can produce a thrust of 0.29 mN at a peak voltage of 15.3 kV, which is adequate for propelling a surface vessel. At this level of thrust, the power consumption is 2.15 W. These results underscore the potential of the DBD thruster as a power source for micro-mechanical devices.

REFERENCES

- [1] N. E. Jewell-Larsen and E. Tran, "Design and optimization of electrostatic fluid accelerators," *IEEE TRANSACTIONS ON DIELECTRICS AND ELECTRICAL INSULATION*, vol. 13, no. 1, pp. 191–203, FEB 2006.
- [2] Y. Gu, X. Cai, K. Thakuri, W. Yang, Y. Guo and W. Li, "Microrobotic Flight Enabled by Ultralight Ion Thrusters with High Thrust-to-Weight Ratio and Low Fabrication Cost," 2024 IEEE International Conference

- on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 6036–6042.
- [3] Q. N. Tao, Z. Chen, Y. Q. Cao, and W. Li, "High-thrust ultralight ion wind engine toward controllable aerial microrobots," *ENERGY*, vol. 339, DEC 1 2025, Art no. 139027.
- [4] Y. Z. Sun, Y. C. Qiu, A. L. Nie, and X. D. Wang, "Experimental research on inactivation of bacteria by using dielectric barrier discharge," *IEEE Trans. Plasma Sci*, vol. 35, no. 5, OCT 2007, pp. 1496–1500.
- [5] N. D. Vaze et al., "Inactivation of bacteria in flight by direct exposure to nonthermal plasma," *IEEE Trans. Plasma Sci*, vol. 38, no. 11, NOV 2010, pp. 3234–3240.
- [6] S. Sato, T. Enokido, and N. Ohnishi, "Fabrication of a multi-stage plasma synthetic jet actuator using printed electronics," *AIP Adv.*, vol. 11, no. 4, APR 2021, Art no. 045105.
- [7] Q. T. Algwari and D. O'Connell, "Plasma Jet Interaction With a Dielectric Surface," *IEEE Transactions on Plasma Science*, vol. 39, no. 11, pp. 2368–2369, 2011.
- [8] G. Uchida, K. Takenaka, K. Kawabata, and Y. Setsuhara, "Influence of He Gas Flow Rate on Optical Emission Characteristics in Atmospheric Dielectric-Barrier-Discharge Plasma Jet," *IEEE Transactions on Plasma Science*, vol. 43, no. 3, pp. 737–744, 2015.
- [9] K. Iranshahi and M. Mani, "Dielectric barrier discharge actuators employed as alternative to conventional high-lift devices," *J. Aircraft*, vol. 55, no. 5, SEP 2018, pp. 2104–2113.
- [10] A. Duchmann, B. Simon, C. Tropea, and S. Grundmann, "Dielectric Barrier Discharge Plasma Actuators for In-Flight Transition Delay," *AIAA JOURNAL*, vol. 52, no. 2, pp. 358–367, FEB 2014.
- [11] P. Warlitz, M. T. Hehner, S. Pasch, J. Serpieri, T. Blank, and J. Kriegseis, "On durable materials for dielectric-barrier discharge plasma actuators," *Sensors and Actuators A: Physical*, vol. 366, 2024.
- [12] F. O. Thomas, T. C. Corke, M. Iqbal, A. Kozlov, and D. Schatzman, "Optimization of Dielectric Barrier Discharge Plasma Actuators for Active Aerodynamic Flow Control," *AIAA JOURNAL*, vol. 47, no. 9, pp. 2169–2178, SEP 2009.
- [13] A. Shaygani and K. Adamiak, "The Dielectric Surface Conductivity Effect on the Dielectric Barrier Discharge Actuator Characteristics," *IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS*, vol. 58, no. 1, pp. 767–775, JAN 2022.
- [14] A. Tang, A. Mamishev, and I. Novosselov, "Multi-electrode dielectric barrier discharge actuators: Geometrical optimization of high power density array," *Sensors and Actuators A: Physical*, vol. 388, 2025.
- [15] F. Rodrigues, A. Mushyam, J. Pascoa, and M. Trancossi, "A new plasma actuator configuration for improved efficiency: the stair-shaped dielectric barrier discharge actuator," *Journal of Physics D: Applied Physics*, vol. 52, no. 38, p. 385201, 2019/07/15 2019.
- [16] S. Grosse and D. Angland, "Parametric investigation of the fluid mechanic performance of an AC dielectric barrier discharge plasma actuator," *JOURNAL OF PHYSICS D-APPLIED PHYSICS*, vol. 53, no. 45, NOV 4 2020, Art no. 455202.
- [17] E. Pescini, L. Francioso, M. G. De Giorgi, and A. Ficarella, "Investigation of a Micro Dielectric Barrier Discharge Plasma Actuator for Regional Aircraft Active Flow Control," *IEEE TRANSACTIONS ON PLASMA SCIENCE*, vol. 43, no. 10, pp. 3668–3680, OCT 2015.
- [18] J. Jolibois and E. Moreau, "Enhancement of the Electromechanical Performances of a Single Dielectric Barrier Discharge Actuator," *IEEE TRANSACTIONS ON DIELECTRICS AND ELECTRICAL INSULATION*, vol. 16, no. 3, pp. 758–767, JUN 2009.
- [19] D. E. Ashpis and M. C. Laun, "Dielectric Barrier Discharge Plasma Actuator Thrust Measurement Methodology Incorporating Antithrust Hypothesis," *AIAA JOURNAL*, vol. 55, no. 12, pp. 4181–4192, DEC 2017.
- [20] M. Wicks and F. O. Thomas, "Effect of Relative Humidity on Dielectric Barrier Discharge Plasma Actuator Body Force," *AIAA JOURNAL*, vol. 53, no. 9, pp. 2801–2804, SEP 2015.
- [21] G. I. Font, C. L. Enloe, J. Y. Newcomb, A. L. Teague, A. R. Vasso, and T. E. McLaughlin, "Effects of Oxygen Content on Dielectric Barrier

- Discharge Plasma Actuator Behavior," *AIAA JOURNAL*, vol. 49, no. 7, pp. 1366–1373, JUL 2011.
- [22] G. P. Vafakos, P. K. Papadopoulos, and P. Svarnas, "A three-stage plasma model based on one-way coupling of plasma dynamics, ionic motion, and fluid flow: Application to DBD plasma actuators," *Journal of Applied Physics*, vol. 137, no. 4, 2025.
 - [23] D. Redchys et al., "Flow Structure Control Using Plasma Actuators," *AIAA Journal*, vol. 62, no. 7, pp. 2548–2561, 2024.
 - [24] K. Nakai, A. Nakano, and H. Nishida, "Validity of Three-Fluid Plasma Modeling for Alternating-Current Dielectric-Barrier-Discharge Plasma Actuator," *AIAA Journal*, vol. 59, no. 4, pp. 1313–1327, 2021.
 - [25] Z. Zuo and X. Zhang, "Flow Control over a Two-Element Airfoil Using a Dielectric-Barrier-Discharge Plasma Actuator," *AIAA Journal*, vol. 63, no. 9, pp. 3571–3581, 2025.
 - [26] S. Mohsenimehr and A. von Keudell, "Surface Dielectric Barrier Discharge (sDBD) for Flow Control in Plasma Conversion," *Plasma Chemistry and Plasma Processing*, vol. 43, no. 6, pp. 1633–1649, 2023/11/01 2023.
 - [27] X. Zhang, J. Zhou, and Z. Ma, "Periodic Vortices Created by Plasma Actuator with Low Frequency," *AIAA Journal*, vol. 62, no. 9, pp. 3436–3447.
 - [28] L. Pang, K. He, D. Di, Q. Zhang, and C. Liu, "Capacitances and energy deposition curve of nanosecond pulse surface dielectric barrier discharge plasma actuator," *Review of Scientific Instruments*, vol. 85, no. 5, 2014.
 - [29] Q. Tao, X. Wang, Y. Gu, W. Li, "An ion discharge-driven thruster based on a lithium niobate piezoelectric transformer," *Micromachines*, vol. 16, no. 3, 2025, Art no. 277.
 - [30] N. L. Mai et al., "Ion-wind powered boat using a novel wire-dielectric water system," *Sensors and Actuators A: Physical*, vol. 383, 2025.
 - [31] A. V. Likhanskii, M. N. Shneider, S. O. Macheret, and R. B. Miles, "Modeling of dielectric barrier discharge plasma actuator in air," *JOURNAL OF APPLIED PHYSICS*, vol. 103, no. 5, MAR 1 2008, Art no. 053305.
 - [32] D. Hua, Y. Shen, T. Zhang, F. Yang, J. Wu, and X. Liu, "A Folding Magnetic Soft Sheet Robot With Real - Time Reconfigurable Magnetization for Targeted Drug Delivery," *SmartBot*, 02/14 2026.
 - [33] X. Yang et al., "Morphing matter: from mechanical principles to robotic applications," *Soft Science*, vol. 3, p. 38, 11/01 2023.
 - [34] R. Jousset, A. Leroy, R. Weber, H. Rabat, S. Loyer, and D. Hong, "Plasma morphology and induced airflow characterization of a DBD actuator with serrated electrode," *JOURNAL OF PHYSICS D-APPLIED PHYSICS*, vol. 46, no. 12, MAR 27 2013, Art no. 125204.
 - [35] E. Moreau, K. Bayoda, and N. Benard, "Streamer propagation and pressure waves produced by a nanosecond pulsed surface sliding discharge: effect of the high-voltage electrode shape," *JOURNAL OF PHYSICS D-APPLIED PHYSICS*, vol. 54, no. 7, FEB 18 2021, Art no. 075207.